Amrita School of Computing, Amritapuri Campus

Department of Computer Science and Engineering

23CSE202 DATABASE MANAGEMENT SYSTEMS • S3 B.Tech CSE

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2. UI Design Document

Group No: C15

Project Title: ShopSphere: A Relational Database Management System for a Multi-Vendor Online Shopping Platform

Team Members:

1. SRAVAN N N (AM.SC.U4CSE25251)
2. HARITH BHUVAN S (AM.SC.U4CSE25222)
3. DEVANG DILEEP (AM.SC.U4CSE25213)
4. JASIMKHAN J L (AM.SC.U4CSE25233)

2.1. Functional requirements:

1. Multi-Role Navigation: Role switching between Customer Storefront, Vendor Dashboard, Admin Portal, and Cart & Atomic Checkout.
2. Product Catalogue & Search: Multi-vendor catalog search with autocomplete, category pills, and multi-criteria filters (category, price
3. Vendor Inventory & Order Fulfillment: Real-time inventory table with low-stock trigger alerts (stock <= threshold) and order fulfillment
4. Platform Admin & DDL Management: Vendor approval workflow (Pending/Approved/Rejected), category taxonomy hierarchy tree, ar
5. Cart, Coupon Engine & Atomic DBMS Checkout: Multi-vendor cart, eligibility coupon validation (SAVE15, SAVE20), and ACID SQL t

2.2. Non-functional requirements:

1. Responsiveness & Layout: Mobile-first flexible CSS Grid & Flexbox layout with left vertical rail sidebar, top search capsule, and botto
2. Design Aesthetic: Dark Glassmorphism (#070a12 canvas, #0b1120 cards, backdrop blur, electric blue/purple gradients, Inter & Outfi
3. Reactivity & Micro-animations: Smooth Framer Motion card lift effects, button micro-feedback, and real-time trigger event logging.
4. Query Performance & Efficiency: Sub-40ms database query response latency with zero main thread blocking.

2.3. List of AI tools used for building UI:

1. Antigravity AI (Google DeepMind) - Architectural design, Next.js SPA development, Supabase relational data modeling, and PDF bu

2.4. Prompt (if any):

1. Tool: Antigravity AI | Prompt: Act as a Principal UI/UX Architect... Build a production-grade Web UI for ShopSphere RDBMS...
2. Tool: Antigravity AI | Prompt: Migrate fully into Next.js with Supabase backend matching attached dark glassmorphism design...
3. Tool: Antigravity AI | Prompt: Convert all product prices, total revenues, subtotal, and order breakdown figures into Indian Rupees (INR).

2.5. List of files:

- app/layout.tsx (Next.js Root Layout wrapper)
- app/page.tsx (ShopSphere Main SPA Page & State Controller)
- app/globals.css (TailwindCSS Directives & Dark Glassmorphism Tokens)
- components/Sidebar.tsx (Left Vertical Icon Rail Sidebar Component)
- components/Header.tsx (Search Capsule Header & Role Switcher)
- components/CategoryPills.tsx (Horizontal Category Selector Pills)
- components/ProductCard.tsx (Multi-Vendor Product Card Component in INR)
- components/RightDrawer.tsx (Recommended For You Side Drawer Component)
- components/BottomDock.tsx (Floating Capsule Bottom Navigation Dock)
- components/VendorDashboard.tsx (Vendor Sales, Inventory & Fulfillment)
- components/AdminPortal.tsx (Admin Vendor Approval, Category Tree & Triggers)
- components/AtomicCheckoutModal.tsx (Multi-Vendor Cart, Coupon & Atomic Checkout)
- components/AddProductModal.tsx (New Product Creation Form Modal)
- lib/supabase.ts (Supabase Client Layer & INR Relational Seed Data)
- supabase/schema.sql (Supabase Production Relational DDL & Triggers Schema)

2.6. UI Screenshot:

[ShopSphere Next.js Single Page Application Dark Glassmorphism Interface]

Figure 2.1: Live Application running at http://localhost:3001 featuring Left Icon Rail, Capsule Search, 4-Column Product Grid in INR, and Floati

2.7. GitHub Link:

GitHub Repository: https://github.com/Sravan-NN/ShopSphere-DBMS